

C language $\Rightarrow$ data types ~~X~~
variables ~~X~~

Conditional ~~statements~~

loops

operators

functions

~~structs~~
~~union~~

pointer

Python $\div$ object oriented program language

Interpreted language

Dynamic

Low level vs high level language

↳ machine understandable language

↳ Binary

↳ mov 10 x

→ high level language → human readable

↳ All of programming language

Compiled vs Interpreted

1) Compiled language

c → gcc file.c ⇒ F7
↳ a.out → F9 ↘
output

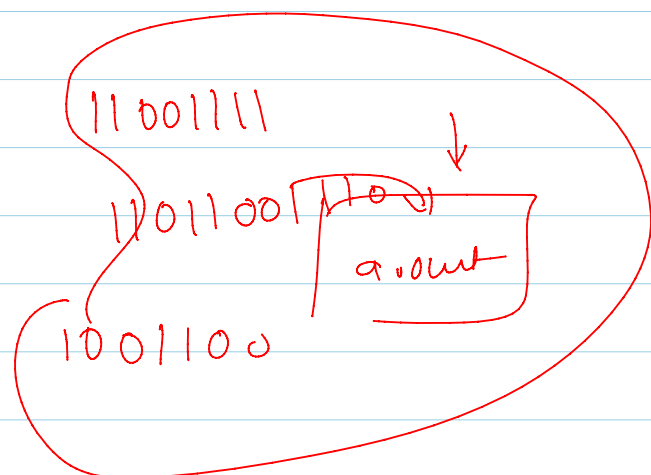
```
#include <stdio.h>;
```

```
int main () { ←
```

```
printf("Hello");
```

```
return -1;
```

```
}
```



`print("Hello")` → 11011011- →

`for i in range(10):`
`print(i)` } → 110111010-1

`print(7/0)` → X

Statically typed / Dynamically typed

Statically typed language

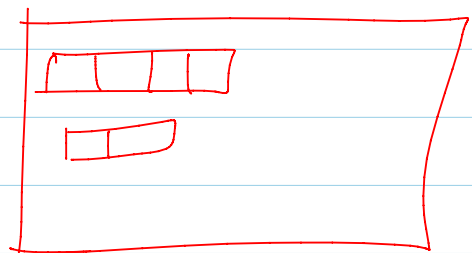
4B

← `int age = 24;`

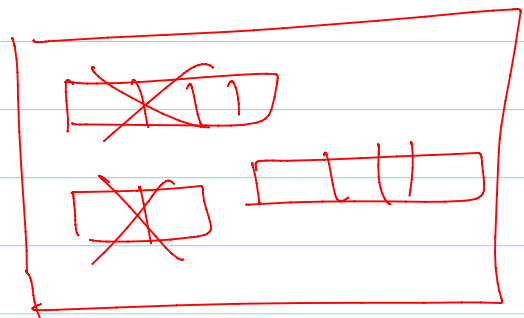
← datatype var-name = data

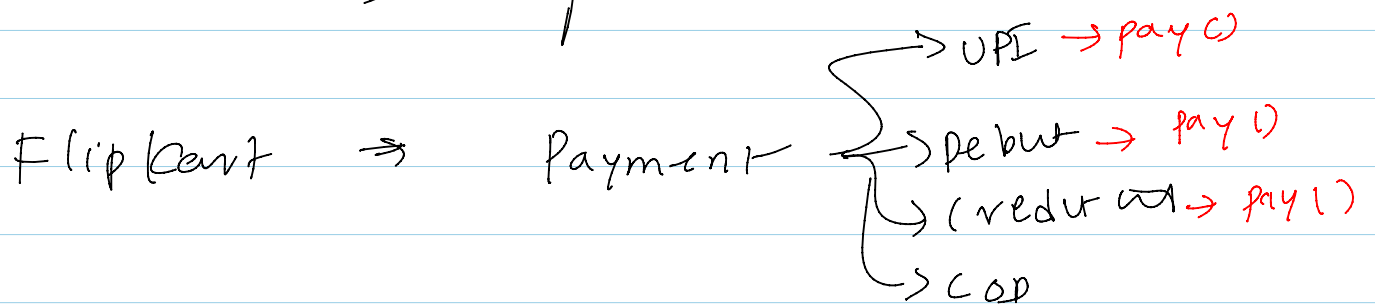
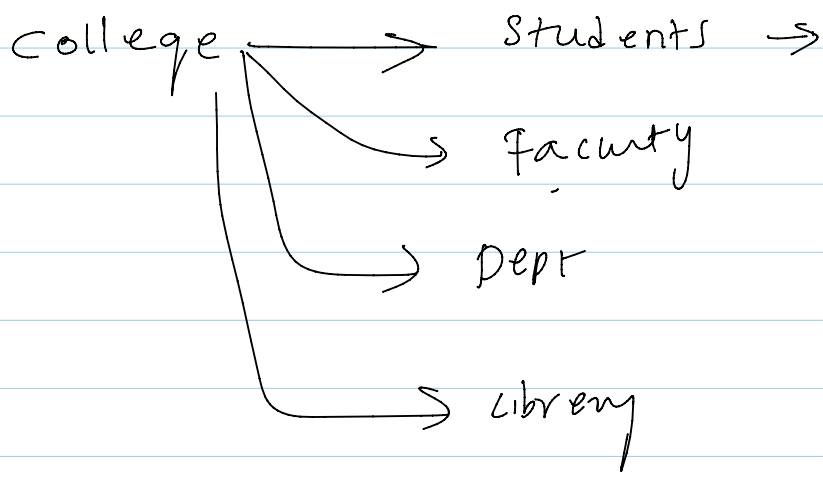
`int age;`

`char gender;`



X `a = 10` ←
`a = 'M'` ←
`a = "paran"` ←





steps to open python file in IDLE

creation

① open files

② select D_drive

③ right click and click on new and create text document

④ rename text document to test.py

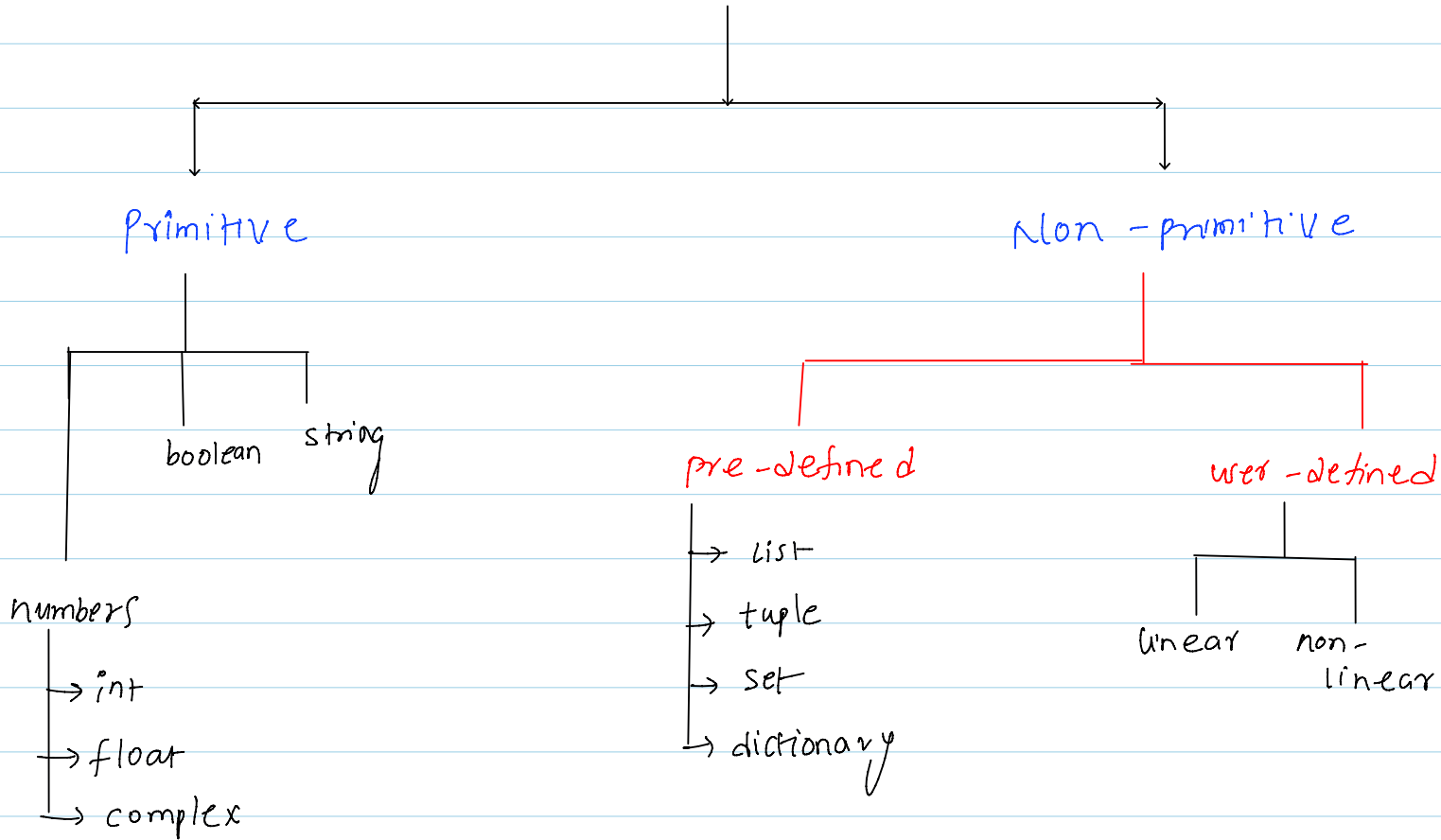
open

⑤ Go to file → click on file → click open
select D drive

⑥ select created file and open

Data types

data types



Variables:

var-name = data

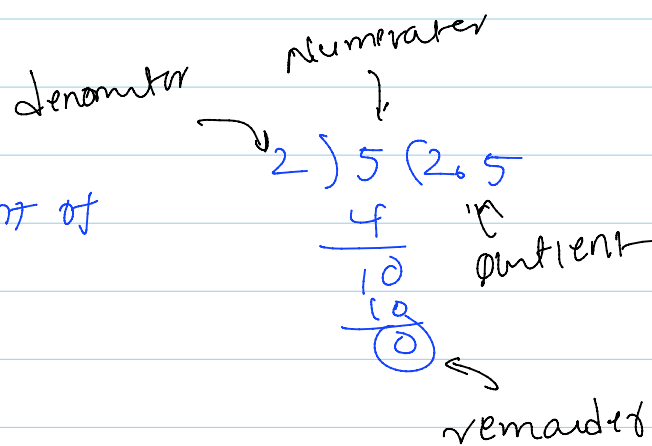
→ a variable will hold some data which will be change in future

→ a variable name should only start with a alphabet (a-z, A-Z) and after that we can keep any character

Operators:

1) Arithmetic: +, -, *, /, //, %, **, floor, modulo, exp

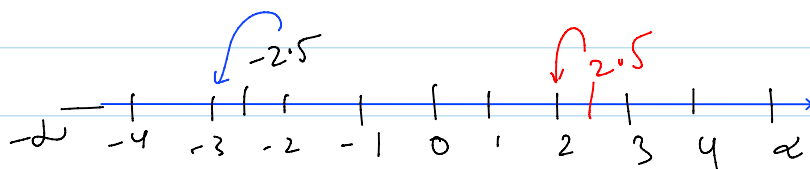
/ → returns the quotient of division



// → returns the division of a/b cut down to nearest and less integer value

$$-5//2 \rightarrow -2.5$$

$$5//2 \rightarrow 2.5$$



$\% \rightarrow$ which will return remainder in a Division

$$10 \% 3 \rightarrow \begin{array}{r} 3 \overline{) 10} \\ \underline{-9} \\ 1 \end{array}$$

$$-10 \% 3 \rightarrow ?$$

$$a \% b = a - (b * \overset{\downarrow}{a // b})$$

$$\begin{aligned} 10 \% 3 &\Rightarrow 10 - (3 * 10 // 3) \\ &10 - (3 * 3) \\ &10 - 9 \Rightarrow (1) \end{aligned}$$

$$\begin{aligned} -10 \% 3 &\Rightarrow -10 - (3 * -10 // 3) \\ &-10 - (3 * -4) \\ &-10 - (-12) \\ &-10 + 12 \Rightarrow (2) \end{aligned}$$

In Java it was truncated $a - (b * \text{trunc}(a/b))$

$$\begin{aligned} 10 / 3 &\rightarrow 3.3\overline{3} \rightarrow (3) \\ -10 / 3 &\rightarrow -3.3\overline{3} \rightarrow (-3) \end{aligned}$$

In python that will be floor

$$\begin{aligned} 10 // 3 &\rightarrow 3 \\ -10 // 3 &\rightarrow -4 \end{aligned}$$

$$a - (b * (a // b))$$

② Assignment operators

$+=$, $-=$, $*=$
 $/=$, $//=$, $./=$
 $**=$,

③ Comparison operators

$(\quad)$
 $=$, $!=$, $<$, $<=$
 $>$, $>=$

→ these return a Boolean value based on a condition b/w two / more variables

④ Logical operators (and , or , not)

→ club any two / more conditional statements to get one final Boolean value

* *

⑤ Bitwise operators $(\quad)$ $\&$, $|$, $\sim$, $\wedge$ $\ll$, $\gg$

→ will work up on bit level to perform operations

* * ⑥ Identity operators (is , is not)

→ checks for memory address equality b/w two variables and returns boolean

⑦ membership operators (in, not in)

→ checks whether an element is in a collection or not and returns boolean value.

$\&, |, \sim, \wedge, <<, >>$

even/odd

$\hookrightarrow a \% 2 \rightarrow$

$1 \rightarrow \text{odd}$

$0 \rightarrow \text{even}$

$10 \rightarrow$

$7 \rightarrow$

A	B	A & B
T	T	T
T	F	F
F	T	F
F	F	F

A	B	A B
T	T	T
T	F	T
F	T	T
F	F	F

A	B	A ^ B
T	T	F
T	F	T
F	T	T
F	F	F

$10 \rightarrow 000010100 \ll 1$

$1010 \rightarrow 10 \times 2$

$10100 \rightarrow 20 \times 2$

$101000 \rightarrow 40 \times 2$

$1010000 \rightarrow 80$
64 32 16 8 4 2 1

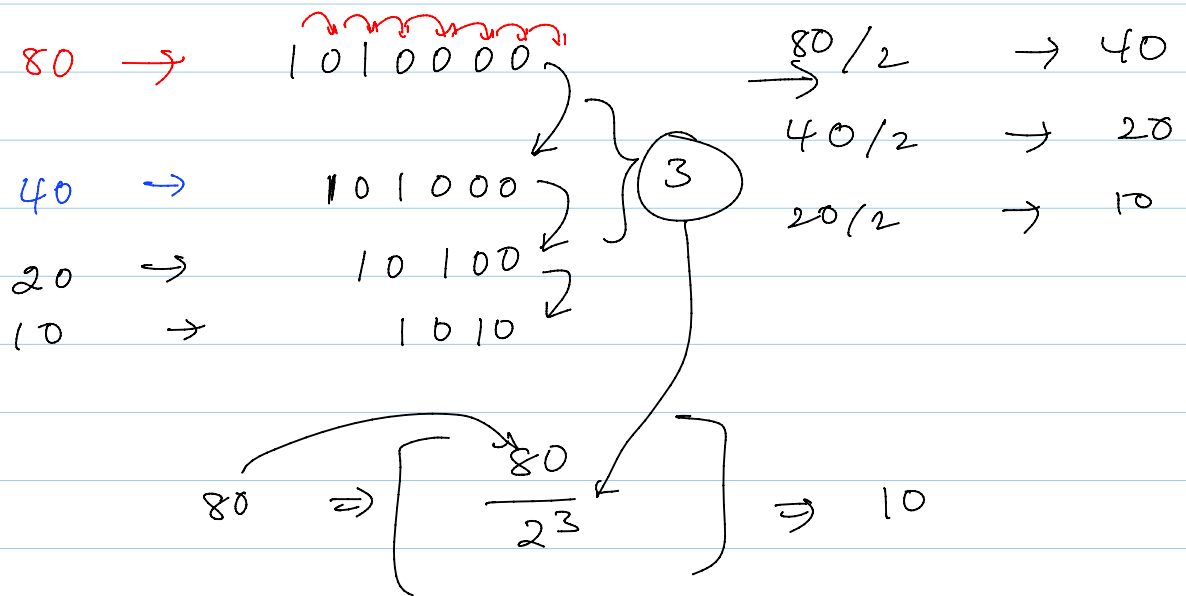
$1 \rightarrow 2 \rightarrow 2^1$

$2 \rightarrow 4 \rightarrow 2^2$

$3 \rightarrow 8 \rightarrow 2^3$

$4 \rightarrow 16 \rightarrow 2^4$

$$(num \ll k) = (num * 2^k)$$



$$\text{num} \gg k = \frac{\text{num}}{2^k}$$

✱ ✱

(XOR) (^) operator
✱

$$\begin{array}{r}
 a = 10 \rightarrow 1010 \\
 a = 10 \rightarrow 1010 \\
 \hline
 0000
 \end{array}$$

$$a \wedge a = 0$$

$$\begin{array}{r}
 a = 10 \rightarrow 1010 \\
 b = 2 \rightarrow 0010 \\
 \hline
 1000 \rightarrow 8
 \end{array}$$

$$a \wedge b = c$$

$$\begin{array}{r}
 8 \rightarrow 1000 \\
 10 \rightarrow 1010 \\
 \hline
 0010 \rightarrow 2
 \end{array}$$

$$10 \wedge 2 = 8$$

$$8 \wedge 10 = 2$$

$$(10 \wedge 2) \wedge 10 = 2$$

$$0 \wedge 2 \Rightarrow 2$$

$$\begin{array}{r}
 1 + 2 + 3 = 6 \\
 \vee \\
 3 + 3 = 6
 \end{array}$$

$$\cancel{temp} = a$$

$$a = b$$

$$b = \cancel{temp}$$

$$\left. \begin{array}{l} a \wedge a = 0 \\ a \wedge 0 = a \\ a \wedge b \wedge a = b \end{array} \right\}$$

$$a = 10$$

$$b = 8$$

$$a = a \wedge b \rightarrow 10 \wedge 8 \rightarrow 2$$

$$a = 2$$

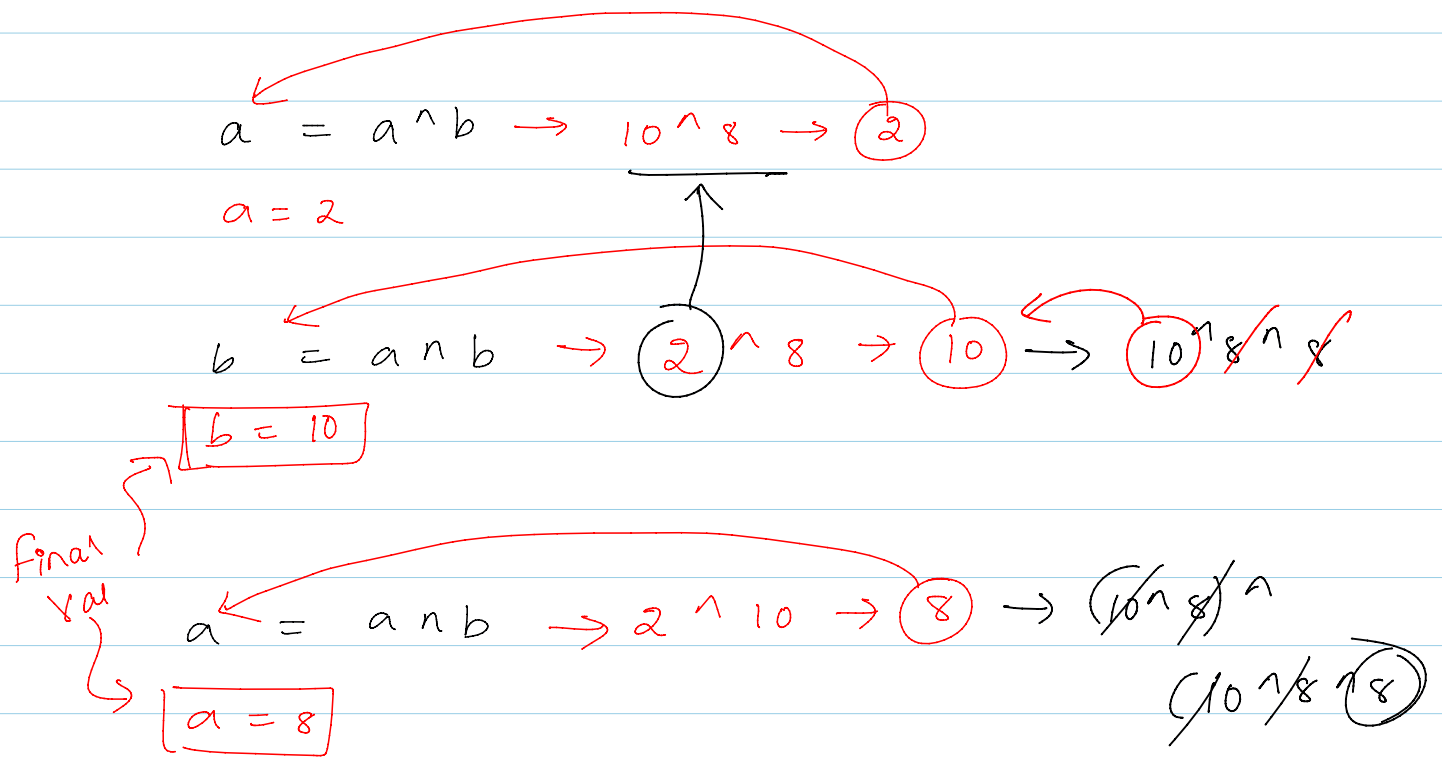
$$b = a \wedge b \rightarrow 2 \wedge 8 \rightarrow 10$$

$$\boxed{b = 10}$$

$$a = a \wedge b \rightarrow 2 \wedge 10 \rightarrow 8$$

$$\boxed{a = 8}$$

final
val



Swapping of two numbers

$a = a \wedge b$
 $b = a \wedge b$
 $c = a \wedge b$

$a = a + b$
 $b = a - b$
 $a = a - b$

$temp = a$
 $a = b$
 $b = temp$

$a, b = b, a \rightarrow \text{python}$
 Specific

Conditional Statements:-

→ if (condition)

→ elif (condition)

→ else

if () → only one

elif ()

elif ()

elif ()

elif ()

elif ()

elif ()

else

→ only one

many permitted

n=10

Ex

→ if (cond) {x}
 else {y}

→ if (^{ev}cond₁) { }
 if (^{od}cond₂) { }

if-else ladder

if (c₁) { }

Kaju Katket

elif (c₂) { }

Jalebi

elif (c₃) { }

Laddu

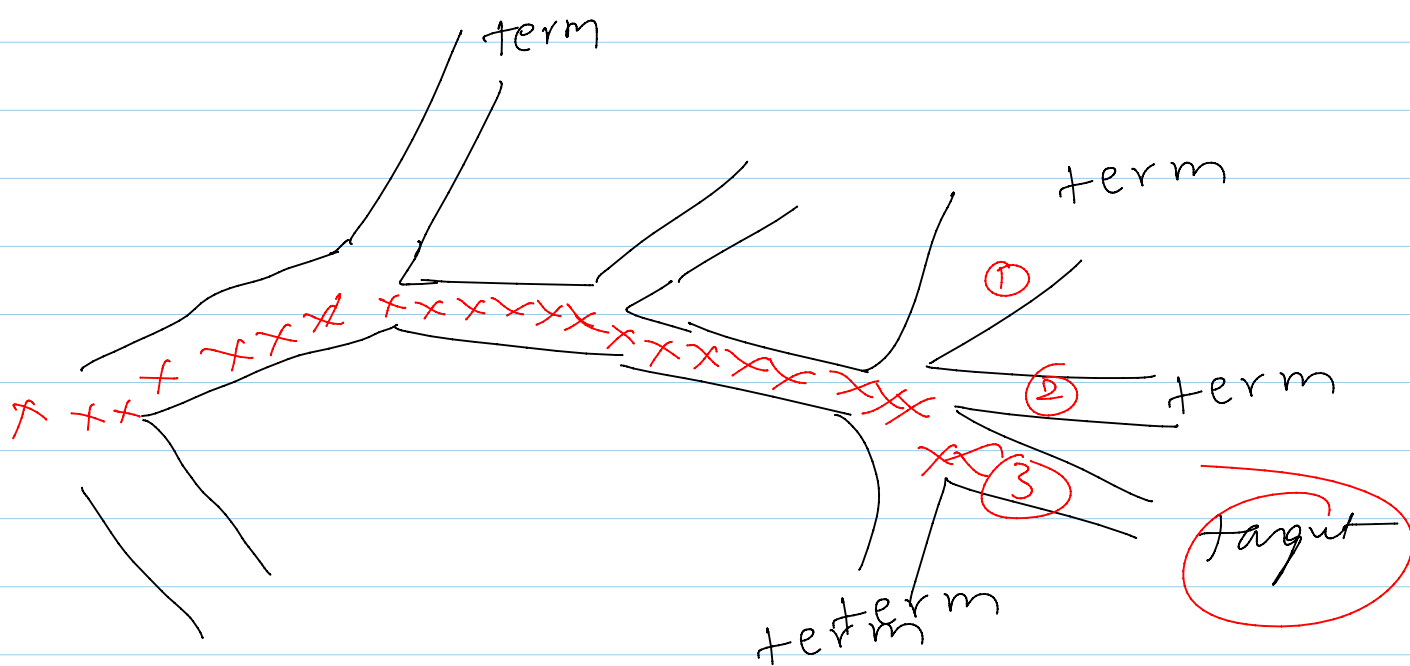
elif (c₄) { }

Mysore Pak

→ else { }

Anything Available

Conditional Statements



if ($n < 0$) -ve
else if ($n = 0$) 0
else 1

